Home / Grade 8 / Geometry / Quarter 2

Pythagoras’ theorem, and different proofs of it

The most famous relation in mathematics, and three ways of seeing why it must be true.

LESSON 20–21 | 2 × 40 MIN | GEOMETRY 8, TEMA 17 | STAGE 9 · 5.5

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State Pythagoras’ theorem precisely.
- Prove it by the square-rearrangement method.
- Find the hypotenuse or a leg from the other two sides.
- Recognise the common Pythagorean triples.

TEXTBOOK

National	Тема 17, pp. 41–43
Cambridge	Learner’s Book pp. 124–126
Workbook	Workbook 5.5

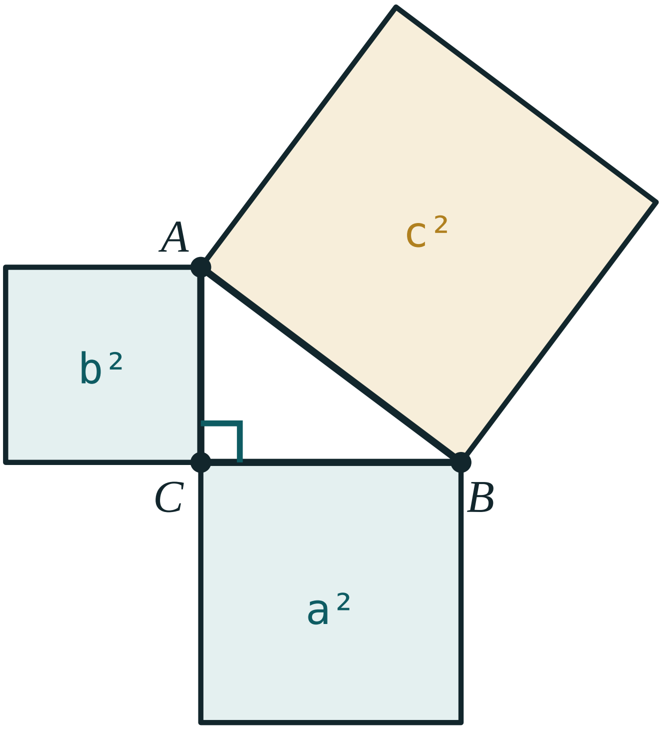
01 Explanation

The theorem

THEOREM

In a right-angled triangle the square of the hypotenuse equals the sum of the squares of the legs:

$$a^2 + b^2 = c^2$$

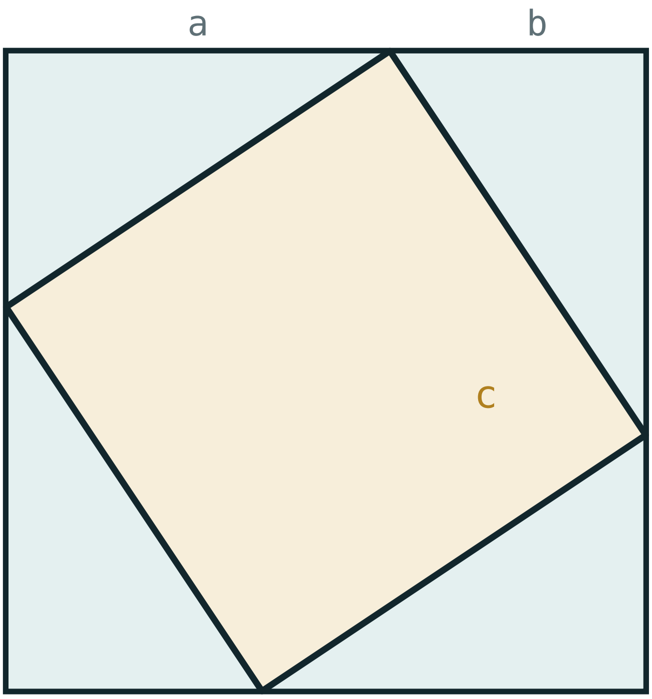


$$a^2 + b^2 = c^2$$

The two smaller squares together have exactly the area of the large one.

The picture is the statement: build a square on each side, and the two small ones together fill the big one exactly.

Proof 1 — the rearrangement



$$(a + b)^2 = 4 \cdot \tfrac{1}{2}ab + c^2$$

Four copies of the triangle inside a square of side a + b. What is left in the middle is a square of side c.

Take a square of side $a + b$ and place four copies of the right triangle inside it, one in each corner. The area can be counted two ways:

$$(a + b)^2 = 4 \cdot \frac{1}{2} ab + c^2$$

$$a^2 + 2ab + b^2 = 2ab + c^2$$

$$a^2 + b^2 = c^2 \quad \blacksquare$$

The middle really is a square: each of its angles is $180° - \alpha - \beta = 180° - 90° = 90°$, and all four sides are hypotenuses of congruent triangles.

Proof 2 — by similar triangles

Drop the height CH from the right angle onto the hypotenuse. It cuts the triangle into two smaller triangles, each similar to the original (they share an acute angle and have a right angle). Comparing sides gives $a^2 = c \cdot BH$ and $b^2 = c \cdot AH$. Adding:

$$a^2 + b^2 = c(BH + AH) = c \cdot c = c^2 \quad \blacksquare$$

This proof is shorter, and it produces two useful formulas on the way.

Using the theorem

To find the **hypotenuse**, add and take the root. To find a **leg**, subtract first:

$$c = \sqrt{a^2 + b^2} \qquad a = \sqrt{c^2 - b^2}$$

TRIPLES WORTH RECOGNISING ON SIGHT

3, 4, 5 · 5, 12, 13 · 8, 15, 17 · 7, 24, 25 · 20, 21, 29 — and every multiple of them, such as *6, 8, 10* and *9, 12, 15*.

THE HYPOTENUSE IS THE LONGEST SIDE

If your answer for c comes out smaller than a leg, you have added where you should have subtracted.

02 Worked examples

EXAMPLE 1 **The legs of a right triangle are 9 cm and 12 cm. Find the hypotenuse.**

1. $c^2 = 9^2 + 12^2$

   Pythagoras.

2. $= 81 + 144 = 225$

3. $c = \sqrt{225} = 15\ cm$

   Recognise 9, 12, 15 — three times 3, 4, 5.

ANSWER

$15\ cm$

EXAMPLE 2 **The hypotenuse is 26 cm and one leg is 10 cm. Find the other leg.**

1. $b^2 = 26^2 - 10^2$

   Looking for a leg, so subtract.

2. $= 676 - 100 = 576$

3. $b = 24\ cm$

   The triple 10, 24, 26 is twice 5, 12, 13.

ANSWER

24 cm

EXAMPLE 3 **A ladder 5 m long leans against a wall with its foot 3 m from the wall. How high does it reach?**

1. The wall, the ground and the ladder form a right triangle.

   The ladder is the hypotenuse.

2. $h^2 = 5^2 - 3^2 = 25 - 9 = 16$

3. $h = 4\ m$

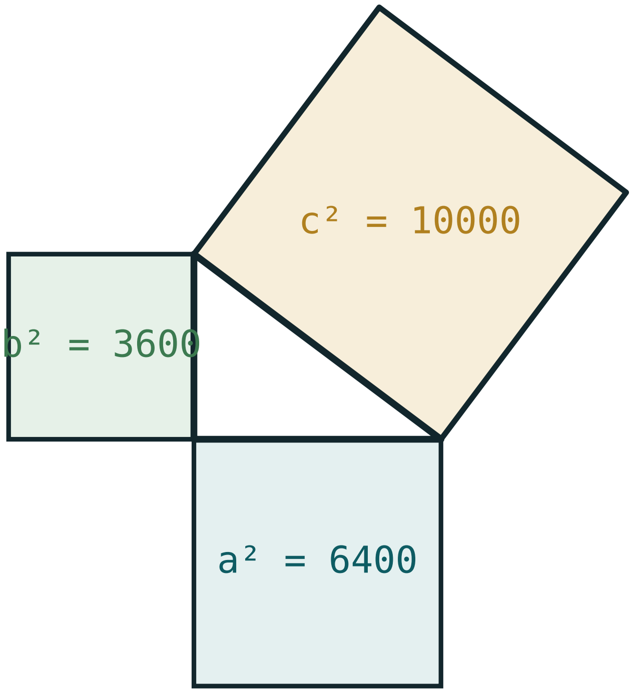
ANSWER

4 m

03 Interactive model

Change both legs and watch a² + b² and c² stay equal to the last digit.

$a^2 + b^2 = c^2$



a **80**	b **60**	c **100**	a^2 **6400**	b^2 **3600**	$a^2 + b^2$ **10000**	c^2 **10000**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Pythagoras’ theorem	Pifagor teoremasi	Теорема Пифагора
Hypotenuse	Gipotenuza	Гипотенуза
Leg (cathetus)	Katet	Катет
Square of a number	Sonning kvadrati	Квадрат числа
Area	Yuza	Площадь
Proof	Isbot	Доказательство
Rearrangement	Qayta joylashtirish	Перекладывание
Pythagorean triple	Pifagor uchligi	Пифагорова тройка
Right angle	To‘g‘ri burchak	Прямой угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Pythagoras' theorem says:

- $a + b = c$
- $a^2 + b^2 = c^2$
- $a^2 - b^2 = c^2$
- $ab = c^2$

Legs 6 and 8. The hypotenuse is:

- *10*
- *14*
- *48*
- *7*

Hypotenuse 13, one leg 5. The other leg is:

- *8*
- *12*
- *18*
- *14*

Which is a Pythagorean triple?

- $2, 3, 4$
- $5, 12, 13$
- $4, 5, 6$
- $6, 7, 8$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Legs 3 and 4. Find the hypotenuse.*

   ANSWER *5*

2. *Legs 6 and 8. Find the hypotenuse.*

   ANSWER *10*

3. *Legs 5 and 12. Find the hypotenuse.*

   ANSWER *13*

4. *Hypotenuse 5, one leg 3. Find the other leg.*

   ANSWER *4*

5. *Hypotenuse 13, one leg 12. Find the other leg.*

   ANSWER *5*

6. *Legs 9 and 12. Find the hypotenuse.*

   ANSWER *15*

7. *Is 3, 4, 5 a Pythagorean triple?*

   ANSWER Yes — $9 + 16 = 25$.

Medium · the standard question

7 PROBLEMS

1. *Legs 8 and 15. Find the hypotenuse.*

   ANSWER *17*

2. *Legs 7 and 24. Find the hypotenuse.*

   ANSWER *25*

3. *Hypotenuse 26, one leg 10. Find the other leg.*

   ANSWER *24*

4. *Hypotenuse 17, one leg 8. Find the other leg.*

   ANSWER *15*

5. *A ladder 5 m long has its foot 3 m from a wall. How high does it reach?*

   ANSWER *4 m*

6. *A rectangle is 9 cm by 12 cm. Find its diagonal.*

   ANSWER *15 cm*

7. *Legs 1 and 1. Find the hypotenuse.*

   ANSWER $\sqrt{2}$

Hard · stretch and reason

7 PROBLEMS

1. *An isosceles triangle has equal sides 13 and base 10. Find its height.*

   ANSWER $h = \sqrt{169 - 25} = 12$

2. *An equilateral triangle has side 6. Find its height.*

   ANSWER $h = \sqrt{36 - 9} = 3\sqrt{3}$

3. *A rhombus has diagonals 16 and 30. Find its side.*

   ANSWER $\sqrt{64 + 225} = 17$

4. *A right triangle has hypotenuse 20 and one leg 12. Find its area.*

   ANSWER other leg 16, area 96

5. *A ladder 13 m long reaches 12 m up a wall. How far is its foot from the wall?*

   ANSWER 5 m

6. *A trapezium has bases 6 and 16 and equal legs 13. Find its height.*

   ANSWER $\frac{16 - 6}{2} = 5$, so $h = 12$

7. *Complete the proof: the middle shape in the rearrangement is a square. Why are its angles 90°?*

   ANSWER Each is $180° - \alpha - \beta$, and $\alpha + \beta = 90°$ in the right triangle.

07 Homework

Homework — 6 problems

Geometry 8, Тема 17, pp. 41–43. Draw a labelled figure for questions 4–6.

1. *Legs 20 and 21. Find the hypotenuse.*
2. *Hypotenuse 25, one leg 7. Find the other leg.*
3. *A rectangle is 8 cm by 15 cm. Find its diagonal.*
4. *An isosceles triangle has equal sides 10 and base 12. Find its height.*

5. *A ladder 10 m long has its foot 6 m from a wall. How high does it reach?*
6. *Write out the rearrangement proof of Pythagoras' theorem with a labelled figure.*